

FAHAD AL HADI

Mechanical Engineer | CAD/CAE & Equipment Design | Advance Manufacturing | R&D

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PROFESSIONAL PROFILE

Mechanical design engineer with hands-on CAD and CAE experience across the full equipment development lifecycle: concept design, gear/mechanism sizing, structural (FEA) validation, GD&T drawing packages, BOMs, and physical prototyping. Delivers non-linear FEA validation of load-bearing components (screw press base frame, topology-optimized bell crank) and complete mechanical designs taken from concept through tested prototype (gravity-powered energy device). Currently completing an MSc in Sustainable Product Creation (Mechanical Engineering) at the University of Luxembourg, building on 3+ years of industrial engineering experience where lean manufacturing, ISO-aligned process improvement, and cross-functional coordination with production, quality, and technical teams were core to the role. Comfortable working under senior engineer guidance, translating requirements into practical mechanical solutions, and following through from design intent to shop-floor feedback. Actively developing Python skills for engineering automation and data analysis.

CORE COMPETENCIES

CAD & 3D Modelling: SolidWorks, Autodesk Inventor, Fusion 360, CATIA

Simulation & CAE: ANSYS (structural & non-linear FEA), Nastran, Topology Optimization, Life Cycle Assessment (LCA)

Additive Manufacturing: Design for AM (DFAM), mesh preparation, build-constraint optimization, topology-driven geometry export

GD&T & Technical Drawings: Dimensioning, tolerancing, manufacturing drawings

Industrial Robotics & Automation: Robot programming, vision-based quality inspection, IIoT integration, sensor & actuator systems

Programming & Data: Python (pandas, matplotlib, automation), MATLAB & Simulink

Manufacturing & Process Engineering: Lean Manufacturing, Six Sigma, Kaizen, 5S, Value Stream Mapping, Root Cause Analysis, ISO 9001 / 14001 / 45001

Tools: MS Office (Excel advanced), Oracle ERP

Languages: English (fluent C1) | French (Beginner actively improving) | Urdu (native)

MECHANICAL DESIGN and ENGINEERING PROJECTS

Complex Engineering Problems

- Gravity-Powered Sustainable Energy Device (Gravity Light) | MSc Project | University of Luxembourg**
 - ✓ Full design-build-test cycle for a gravity-powered sustainable energy device, covering concept development and gear calculations, CAD modelling, GD&T technical drawings, BOM, physical prototyping, and mechanical validation testing against design and end user requirements.
 - ✓ Converted a **12 kg / 1800 mm** descending mass at **0.083 RPM** into **227 RPM** output via a 2734.7:1 gear ratio.
 - ✓ Validated the module 1.5, 172.5 mm planetary ring gear design using Lewis bending equations, keeping the peak stress at **14.47 MPa**, which is well within the 40 to 60 MPa tensile strength of the 3D printed PLA components.
 - ✓ Built and tested a working prototype achieving **1200 RPM / 2 V** output from the drive motor; diagnosed the transmission-efficiency gap versus the 20-minute illumination target and defined a clear optimization path (pulley resizing, ratio reduction, belt/chain transmission) through experimental root-cause analysis.
- Single-Column Screw Press — Structural Validation (20 kN) | MSc Project | University of Luxembourg**
 - ✓ Performed non-linear FEA of a 20 kN-rated screw press base frame (Cast Iron Grade 40, ~31 kg), validating a safety factor of **1.5** with maximum Von Mises stress kept below **200 MPa** and peak displacement of ~1 mm under full load.
 - ✓ Identified stress concentrations at frame corners via FEA and resolved them through fillet redesign, confirming structural integrity without added mass.
- Topology Optimization For Additive Manufacturing | MSc Project | University of Luxembourg**
 - ✓ Redesigned a bell crank component from a **77.1 × 112.3 × 15 mm** design space under a **20 kN** load, through mesh generation, iterative topology optimization, and DfAM-compliant CAD reconstruction.
 - ✓ Validated the optimized geometry (LUVOCOM® 3F PET CF 9780 BK) via non-linear structural and buckling FEA in Nastran, confirming a maximum displacement of **0.374 mm**, Von Mises stress of **74.7 MPa**, and a buckling safety margin of **5.4×** the applied load.
- Reverse Engineering Connecting Rod Cap | MSc Project | University of Luxembourg**
 - ✓ Connecting Rod Cap reconstructed from 3D scan data acquisition through point cloud processing to a fully parametric CAD model and dimensioned manufacturing drawing.

Analysis And Reporting

- Advanced Control - DC Motor Servo System | MSc Project | University of Luxembourg**
 - ✓ Modeled and compared P, I, D, PI, PD, and PID control strategies for a DC motor servo, identifying PID as the optimal controller with ~5% overshoot and zero steady-state error versus 15–25% overshoot for single-term controllers
 - ✓ Assessed the impact of a 0.01 s tachometric sensor delay on closed-loop stability, showing PID maintained the most robust performance across speed, precision, and disturbance rejection.
- Advanced Engineering Materials - Thermal Protection Systems for Orbital Re-entry | MSc Project | University of Luxembourg**
 - ✓ In-depth study of high-performance composite and ceramic materials used in space shuttle TPS and developed understanding of Fiber architecture, material anisotropy, and the interplay between manufacturing process and structural/thermal performance.
- HSE Risk Assessment of a Bata Shoes Manufacturing Facility | BSc Project | U.E.T Lahore, Pakistan**
 - ✓ Identifying hazards, rating risks using OSHA standards, and documenting mitigation recommendations in a structured audit report.
- Operational Improvement Analysis for a Covid-Impacted Food Business | BSc Project | U.E.T Lahore, Pakistan**
 - ✓ Modelling process changes and propose a strategy that projected a 15% profit uplift.

CERTIFICATIONS & PROFESSIONAL TRAINING

Industrial Robotics and Automation 3-Course Certification (88 hrs.) | Digital Learning Hub Luxembourg | Nov- Dec 2025

- From Code to Motion Combining Robotics and Coding for Industrial Automation (40 hrs.)
✓ Python for robotics, motion control, IIoT integration.
- Vision Systems for Industrial Robots (24 hrs.)
✓ Object recognition, vision-based quality inspection, sensor integration.
- Advanced Industrial Robotics (24 hrs.)
✓ Advanced control algorithms, actuator systems, real-world manufacturing applications.

WORK EXPERIENCE

Subject Matter Expert (Domain: Operational Excellence) | MNBEC / The HAWKS Lahore, Pakistan | Mar 22 – Jun 24

- Led business process reengineering and benchmarking engagements for manufacturing clients including Raniyal Chemicals Pvt. Ltd., developing strategic road maps in collaboration with senior consultants.
- Designed a Factory Assessment Tool aligned to ISO 9001:2015, ISO 14001:2015, and ISO 45001:2018 for cross-functional operational assessments across client organizations.
- Implemented a visual management system at Canon Pvt. Ltd. that improved operational efficiency by 30%, reduced changeover delays, and enabled real-time WIP control across production lines.
- Worked with principal consultants and stakeholders in defining the scope of the projects and team building.

Executive Industrial Engineer | Style Textile Pvt. Ltd Lahore, Pakistan | Oct 2020 – July 2022

- Improved plant efficiency from 45% to 75% for ADIDAS operations by deploying lean tools, optimizing line balance, and eliminating unnecessary changeovers.
- Built a workforce skill matrix enabling 80+ workers to perform multi-role tasks, significantly increasing operational flexibility across production lines.
- Designed work aids and production templates to reduce dependency on skilled labor, accelerating onboarding and reducing training time for new workers.
- Collaborated with the quality team to deploy an inspection framework that reduced defect rates and improved Mean Time to Repair (MTTR).
- Conducted internal audits, production forecasting, and resource optimization planning for ADIDAS supply chain operations.

Lubrication Performance Study 3 Months Industrial Placement | TotalEnergies Lube House Lahore, Pakistan

- Analyzed the effect of lubrication types on equipment reliability and produced technical findings report documenting performance outcomes.

EDUCATION

MSc in Sustainable Product Creation (Mechanical Engineering) | University of Luxembourg | Sept 2024 – Present

BSc in Industrial & Manufacturing Engineering | University of Engineering and Technology Lahore (U.E.T) | Oct 2016 – Aug 2020

Thesis: Synthesis and characterization of micro/nano aluminum particles through electric discharge machining.

VOLUNTEERING

Founder | One Nation Action Plan 2020 (ONAP 2020) Sahiwal, Pakistan |Mar – Oct 2020

- Founded and led a community relief program during COVID-19, raising 500,000 PKR and delivering one month of grocery support to 80 affected families while protecting the anonymity of both donors and recipients.